Subqueries in SQL Server

1. What is a Subquery?

A **Subquery** is a **query inside another SQL query**.

It is also called:

- **Inner Query** – the query inside
- **Outer Query** – the main query

The **inner query executes first**, and its result is used by the **outer query**.

Basic Syntax

```
SELECT column_name
FROM table_name
WHERE column_name operator
    (SELECT column_name FROM table_name WHERE condition);
```

Execution Flow:

1. Subquery runs first
2. Result is returned to main query
3. Main query uses that result

---

2. Sample Tables (Banking Example)

Let's use simple **Banking tables**.

Customers Table

```
Customers
---------
CustomerID
CustomerName
```

City

Accounts Table

Accounts

AccountID

CustomerID

Balance

AccountType

Transactions Table

Transactions

TransactionID

AccountID

Amount

TransactionType

---

3. Simple Subquery Example

Find customers who have an account

```
SELECT CustomerName
FROM Customers
WHERE CustomerID IN
    (SELECT CustomerID FROM Accounts);
```

How it works

Subquery executes first

```
SELECT CustomerID FROM Accounts
```

Example Result

101

102

103

Outer query becomes

```
SELECT CustomerName
FROM Customers
WHERE CustomerID IN (101,102,103);
```

Result → Customers having accounts.

---

4. Subquery with Comparison Operators

Operators used:

=

>

<

>=

<=

<>

Example

Find accounts having **balance greater than average balance**

```
SELECT AccountID, Balance
FROM Accounts
WHERE Balance >
    (SELECT AVG(Balance) FROM Accounts);
```

Step 1

```
SELECT AVG(Balance) FROM Accounts
```

Result

45000

Step 2

```
SELECT AccountID, Balance
FROM Accounts
WHERE Balance > 45000
```

---

5. Subquery with IN Operator

IN checks multiple values.

Example

Find customers who have accounts.

```
SELECT CustomerName
FROM Customers
WHERE CustomerID IN
    (SELECT CustomerID FROM Accounts);
```

Meaning:

CustomerID exists in Accounts table

---

6. Subquery with EXISTS

EXISTS checks whether the subquery **returns rows**.

Example

Find customers who have accounts.

```
SELECT CustomerName
FROM Customers c
WHERE EXISTS
(
  SELECT 1
```

```
    FROM Accounts a
    WHERE a.CustomerID = c.CustomerID
);
```

Explanation:

For each customer, SQL checks:

Is there any account with same CustomerID?

If yes → customer returned.

---

7. Subquery with ANY

ANY means **compare with any value returned by subquery**.

Example

Find accounts with balance greater than **any savings account balance**

```
SELECT AccountID, Balance
FROM Accounts
WHERE Balance > ANY
(
    SELECT Balance
    FROM Accounts
    WHERE AccountType = 'Savings'
);
```

Meaning:

Balance > at least one savings account balance

---

8. Subquery with ALL

ALL means condition must satisfy **all values**.

Example

Find accounts with balance greater than **all savings accounts**

```
SELECT AccountID, Balance
FROM Accounts
WHERE Balance > ALL
(
   SELECT Balance
   FROM Accounts
   WHERE AccountType = 'Savings'
);
```

Meaning:

Balance greater than highest savings account balance

---

9. Subquery in SELECT Clause

Subquery can also appear in **SELECT column list**.

Example

Show account with **average balance**

```
SELECT
AccountID,
Balance,
(SELECT AVG(Balance) FROM Accounts) AS AvgBalance
FROM Accounts;
```

Result

AccountID	Balance	AvgBalance
1	50000	42000
2	30000	42000

---

10. Subquery in FROM Clause (Derived Table)

A subquery can behave like a **temporary table**.

Example

Find accounts with balance greater than average.

```
SELECT *
FROM
(
   SELECT AccountID, Balance
   FROM Accounts
) AS AccountData
WHERE Balance > 40000;
```

---

11. Correlated Subquery

A **Correlated Subquery** depends on the **outer query row**.

It executes **once for each row** of the outer query.

Example

Find customers whose account balance is above **their own average transaction amount**.

```
SELECT CustomerID
FROM Accounts a
WHERE Balance >
(
```

```
    SELECT AVG(Amount)
    FROM Transactions t
    WHERE t.AccountID = a.AccountID
);
```

Here:

t.AccountID = a.AccountID

The subquery depends on the outer query.

---

12. Subquery vs Join

Subquery	**Join**
Query inside query	Combine tables
Sometimes slower	Often faster
Easy to write	Better performance

Example using **JOIN instead of subquery**

Subquery:

```
SELECT CustomerName
FROM Customers
WHERE CustomerID IN
    (SELECT CustomerID FROM Accounts);
```

Join version:

```
SELECT DISTINCT c.CustomerName
FROM Customers c
JOIN Accounts a
ON c.CustomerID = a.CustomerID;
```

---

13. Types of Subqueries

1️⃣ Single Row Subquery

Returns **one value**

Example

```
SELECT *
FROM Accounts
WHERE Balance >
    (SELECT AVG(Balance) FROM Accounts);
```

2️⃣ Multiple Row Subquery

Returns **multiple values**

Example

```
SELECT *
FROM Customers
WHERE CustomerID IN
    (SELECT CustomerID FROM Accounts);
```

3️⃣ Correlated Subquery

Depends on **outer query row**

Example

```
SELECT *
FROM Accounts a
WHERE Balance >
(
  SELECT AVG(Balance)
  FROM Accounts
```

```
    WHERE CustomerID = a.CustomerID
);
```